Tae Jun Ham

Senior Staff Software Engineer & Tech Lead @ Google

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About

I am a **Senior Staff Software Engineer and Tech Lead at Google**, focusing on **hardware/software performance, resource efficiency, and co-design** for Google's datacenter fleet and distributed systems infrastructure. Specializing in **cross-stack optimization**, my work bridges distributed systems and silicon microarchitecture to untangle complex system behaviors, optimize foundational infrastructure, and guide future SoC designs.

Prior to joining Google, I was a postdoctoral researcher at Seoul National University. I received my **Master's and Ph.D. in Electrical Engineering from Princeton University** under the supervision of Professor Margaret Martonosi, and a B.S.E. degree from Duke University. My research centers on hardware-software co-design for emerging applications and data access optimizations across systems and architectures.

Throughout my career, I have published **30+ works** in top computer architecture and systems venues (ISCA, ASPLOS, MICRO, HPCA, ATC, MLSys, etc.). My research emphasizes architectural acceleration and co-design, with key highly-cited contributions including early Transformer/Attention accelerators and a graph analytics accelerator. My work has been recognized with the **Best Paper Award at MICRO-49**, **IEEE Micro Top Picks** (including Honorable Mention), a **Best Paper Award Nomination at ISPASS**, and the **Samsung Scholarship**. I also actively serve on program committees for leading conferences in the field.

Education

Princeton University, Princeton, NJ USA — **Sep 2012 – Jun 2018**
School of Engineering and Applied Science

- **Ph.D** and **M.A** in Electrical Engineering
- **Dissertation**: Data Access Optimization in Accelerator-oriented Heterogeneous Architecture through Decoupling and Memory Hierarchy Specialization
- **Advisor**: Professor Margaret Martonosi and Professor Juan Luis Aragon

Duke University, Durham, North Carolina USA — **Aug 2009 – Dec 2011**
Pratt School of Engineering

- **B.S.E** in Electrical and Computer Engineering (GPA : 3.95/4.00)
- *Summa Cum Laude, with Distinction in Electrical and Computer Engineering*

Experience

Google, Sunnyvale, California, USA — **Sep 2021 – Present**
Senior Staff Software Engineer & Tech Lead

Lead technical strategy for cross-stack performance, hardware/software co-design, and resource efficiency across Google's datacenter fleet and distributed systems. Previously was a tech lead manager for a ~20-person team before transitioning to a strategic technical lead role.

- **Datacenter Efficiency:** Led fleet-wide optimization initiatives across compute, memory, and storage, delivering significant capacity savings.
- **Hardware/Software Co-Design:** Bridge distributed systems and silicon microarchitecture to evaluate, optimize, and guide the architectural design of future SoC and server platforms.
- **ML for Systems:** Focus on leveraging machine learning to drive system efficiency and performance, and unlock new opportunities in hardware/software co-design.

Seoul National University, Seoul, Republic of Korea — **Jul 2018 – Jul 2021**
Postdoctoral Researcher

Focused on Computer Architecture and Systems research.

- **Supervisor:** Jae W. Lee.

- *(Note: My position at Seoul National University also fulfilled my mandatory military service duty)*

Microsoft Research, Cambridge, UK **May – Aug 2016**
Graduate Research Intern

Investigated and designed an efficient secure memory architecture featuring near-data computation.

- **Collaborator:** Stavros Volos.

Intel Labs — Parallel Computing Lab, Santa Clara, USA **May – Nov 2015**
Graduate Technical Intern

Conducted research and development on a custom hardware accelerator tailored for graph analytics.

- **Collaborator:** Lisa Wu Wills.

AMD Research, Austin, USA **Jun – Aug 2013**
Co-op Engineer

Explored heterogeneous memory systems, focusing specifically on stacked DRAM architectures and performance optimization.

- **Collaborator:** Joseph L. Greathouse.

Samsung Advanced Institute of Technology, Yongin, Republic of Korea **Jun – Aug 2012**
Research Intern

Investigated GPU branch divergence problems and proposed architectural solutions to improve execution efficiency.

- **Collaborator:** Yeon-Gon Cho.

Duke University — Systems Architecture Integration Lab, Durham, USA **Jan – May 2012**
Research Assistant

Research on efficient control and management of the heterogeneous memory system.

- **Advisor:** Benjamin C. Lee.

Honors and Awards

- **IEEE MICRO Top Picks (2021)**: Genesis paper selected as one of top 12 computer architecture papers of 2020
- **IEEE MICRO Top Picks Honorable Mention (2021)**: Graphene paper selected as one of top 24 computer architecture papers of 2020
- **ISPASS Best Paper Award Nominee (2020)**: MosaicSim paper
- **MICRO-49 Best Paper Award (2016)**: Graphicionado paper
- **IEEE MICRO Top Picks Honorable Mention (2016)**: DeSC paper (Top 23 computer architecture papers of 2015)
- **Facebook Graduate Fellowship Finalist (2016-2017)**, Facebook, Inc.
- **Gordon Y.S. Wu Fellowship (2012-2017)**, Princeton University
- **Samsung Scholarship (2012-2017)**, Up to $50,000 per year for five years of graduate studies
- **Summa Cum Laude (2011)**, Duke University

Selected Publications

[EMNLP '25] Reliable and Cost-Effective Exploratory Data Analysis via Graph-Guided RAG
Mossad Helali, Yutai Luo, **Tae Jun Ham**, Jim Plotts, Ashwin Chaugule, Jichuan Chang, Parthasarathy Ranganathan, Essam Mansour
*The 2025 Conference on Empirical Methods in Natural Language Processing (**EMNLP**)*

[SAC '25] SkipLSM: Fast Retrieval of Hot Key-Value Pairs on LSM Tree
Jongsung Lee, Sam Son, Jonghyun Bae, Yunho Jin, **Tae Jun Ham**, Jae W. Lee
*The 40th ACM/SIGAPP Symposium on Applied Computing (**SAC**)*
Acceptance Rate : TBD

[ACM TOS '24] An LSM tree augmented with b+ tree on nonvolatile memory
Donguk Kim, Jongsung Lee, Keun Soo Lim, Jun Heo, **Tae Jun Ham**, Jae W. Lee
*ACM Transactions on Storage (**TOS**)*

[ISCA@50 Retrospective] Genesis: A Hardware Acceleration Framework for Genomic Data Analysis
Lisa Wu Wills, **Tae Jun Ham**, Jae W. Lee, Krste Asanovic
The 50th Annual IEEE/ACM International Symposium on Computer Architecture (Retrospective)
98 papers were selected out of 1000+ papers across 25-years of ISCA history

[HPCA '22] Mithril: Cooperative Row Hammer Protection on Commodity DRAM Leveraging Managed Refresh
Michael Jaemin Kim, Jaehyun Park, Yeonhong Park, Wanju Doh, Namhoon Kim, **Tae Jun Ham**, Jae W. Lee, Jung Ho Ahn
*The 28th IEEE International Symposium on High-Performance Computer Architecture (**HPCA**)*
Acceptance Rate : 80/262 ≈ 30%

[ACM TECS '22] MaPHeA: A framework for lightweight memory hierarchy-aware profile-guided heap allocation
Deok-Jae Oh, Yaebin Moon, Do Kyu Ham, **Tae Jun Ham**, Yongjun Park, Jae W. Lee, Jung Ho Ahn, Eojin Lee
*ACM Transactions on Embedded Computing Systems (**TECS**)*

[MLSys '22] ULPPACK: Fast Sub-8-bit Matrix Multiply on Commodity SIMD Hardware
Jaeyeon Won, Jeyeon Si, Sam Son, **Tae Jun Ham**, Jae W. Lee
*Proceedings of Machine Learning and Systems (**MLSys**)*

[HPCA '22] ANNA: Specialized Architecture for Approximate Nearest Neighbor Search
Yejin Lee, Hyunji Choi, Sunhong Min, Hyunseung Lee, Sangwon Baek, Dawoon Jeong, Jae W. Lee, **Tae Jun Ham**
*The 28th IEEE International Symposium on High-Performance Computer Architecture (**HPCA**)*
Acceptance Rate : 80/262 ≈ 30%

[ECCV '22] L3: Accelerator-Friendly Lossless Image Format for High-Resolution, High-Throughput DNN Training
Jonghyun Bae, Woohyeon Baek, **Tae Jun Ham**, Jae W. Lee
*European Conference on Computer Vision (**ECCV**)*
Acceptance Rate : 1650/5803 ≈ 28%

[IEEE TC '22] Architecting a Flash-based Storage System for Low-cost Inference of Extreme-scale DNNs
Yunho Jin, Shine Kim, **Tae Jun Ham**, Jae W. Lee
*IEEE Transactions on Computers (**TC**)*

[ASPLOS '21] MERCI: Efficient Embedding Reduction on Commodity Hardware via Sub-Query Memoization
Yejin Lee, Seong Hoon Seo, Hyunji Choi, Hyoung Uk Sul, Soosung Kim, Jae W. Lee, **Tae Jun Ham**
*The 26th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**)*
Acceptance Rate : 75/398 ≈ 19%

[FAST '21] FlashNeuron: SSD-Enabled Large-Batch Training of Very Deep Neural Networks
Jonghyun Bae, Jongsung Lee, Yunho Jin, Sam Son, Shine Kim, Hakbeom Jang, **Tae Jun Ham**, Jae W. Lee
*USENIX Conference on File and Storage Technologies (**FAST**)*
Acceptance Rate : 28/130 ≈ 21%

[FAST '21] Behemoth: A Flash-centric Training Accelerator for Extreme-scale DNNs
Shine Kim, Yunho Jin, Gina Sohn, Jonghyun Bae, **Tae Jun Ham**, Jae W. Lee
*USENIX Conference on File and Storage Technologies (**FAST**)*
Acceptance Rate : 28/130 ≈ 21%

[HPCA '21] Layerweaver: Maximizing Resource Utilization of Neural Processing Units via Layer-Wise Scheduling
Young H. Oh, Seonghak Kim, Yunho Jin, Sam Son, Jonghyun Bae, Jongsung Lee, Yeonhong Park, Dong Uk Kim, **Tae Jun Ham**, Jae W. Lee
*The 27th IEEE International Symposium on High-Performance Computer Architecture (**HPCA**)*
Acceptance Rate : 63/258 ≈ 24%

[IEEE Micro Top Picks '21] Accelerating Genomic Data Analytics with Composable Hardware Acceleration Framework
Tae Jun Ham, David Bruns-Smith, Brendan Sweeney, Yejin Lee, Seong Hoon Seo, U Gyeong Song, Young H. Oh, Krste Asanovic, Jae W. Lee, Lisa Wu Wills
IEEE Micro Special Issue on Top Picks from the 2020 Computer Architecture Conferences

[ISCA '21] ELSA: Hardware-Software Co-design for Efficient, Lightweight Self-Attention Mechanism in Neural Networks

Tae Jun Ham, Yejin Lee, Seong Hoon Seo, Soosung Kim, Hyunji Choi, Sung Jun Jung, Jae W. Lee
The 48th ACM/IEEE International Symposium on Computer Architecture ***(ISCA)***
Acceptance Rate : 76/406 ≈ 19%

[ISCA '21] BOSS: Bandwidth-Optimized Search Accelerator for Storage-Class Memory
Jun Heo, Seung Yul Lee, Sunhong Min, Yeonhong Park, Sung Jun Jung, **Tae Jun Ham**, Jae W. Lee
The 48th ACM/IEEE International Symposium on Computer Architecture ***(ISCA)***
Acceptance Rate : 76/406 ≈ 19%

[ATC '21] ASAP: Fast Mobile Application Switch via Adaptive Prepaging
Sam Son, Seung Yul Lee, Yunho Jin, Jonghyun Bae, Jinkyu Jeong, **Tae Jun Ham**, Jae W. Lee, Hongil Yoon
USENIX Annual Technical Conference ***(ATC)***
Acceptance Rate : 64/341 ≈ 19%

[LCTES '21] MaPHeA: A Lightweight Memory Hierarchy-aware Profile-guided Heap Allocation
Deok-Jae Oh, Yaebin Moon, Eojin Lee, **Tae Jun Ham**, Jae W. Lee, Jung Ho Ahn
ACM SIGPLAN/SIGBED Conference on Languages, Compilers, Tools and Theory of Embedded Systems ***(LCTES)***
Acceptance Rate : 15/37 ≈ 40%

[ICCAD '20] Unlocking Wordline-Level Parallelism for Fast Inference on RRAM-Based DNN Accelerator
Yeonhong Park, Seung Yul Lee, Hoon Shin, Jun Heo, **Tae Jun Ham**, Jae W. Lee
The 39th IEEE/ACM International Conference on Computer-Aided Design ***(ICCAD)***
Acceptance Rate : 127/470 ≈ 27%

[MICRO '20] Graphene: Strong yet Lightweight Row Hammer Protection
Yeonhong Park, Woosuk Kwon, Eojin Lee, **Tae Jun Ham**, Jung Ho Ahn, Jae W. Lee
The 53rd IEEE/ACM International Symposium on Microarchitecture ***(MICRO)***
Acceptance Rate : 82/424 ≈ 19% (• **IEEE Micro Top Picks Honorable Mention**)

[ISCA '20] Genesis: A Hardware Acceleration Framework for Genomic Data Analysis
Tae Jun Ham, David Bruns-Smith, Brendan Sweeney, Yejin Lee, Seong Hoon Seo, U Gyeong Song, Young H. Oh, Krste Asanovic, Jae W. Lee, Lisa Wu
The 47th ACM/IEEE International Symposium on Computer Architecture ***(ISCA)***
Acceptance Rate : 77/428 ≈ 18% (• **IEEE Micro Top Picks**)

[ISCA '20] A Case for Hardware-Based Demand Paging
Gyusun Lee*, Wenjing Jin*, Wonsuk Song, Jeonghun Gong, Jonghyun Bae, **Tae Jun Ham**, Jae W. Lee, Jinkyu Jeong
The 47th ACM/IEEE International Symposium on Computer Architecture ***(ISCA)***
Acceptance Rate : 77/428 ≈ 18% (*Two authors contributed equally.)

[ISCA '20] A Specialized Architecture for Object Serialization with Applications to Big Data Analytics
Jaeyoung Jang, Sung Jun Jung, Sunmin Jeong, Jun Heo, Hoon Shin, **Tae Jun Ham**, Jae W. Lee
The 47th ACM/IEEE International Symposium on Computer Architecture ***(ISCA)***
Acceptance Rate : 77/428 ≈ 18%

[ISPASS '20] MosaicSim: A Lightweight, Modular Simulator for Heterogeneous Systems
Opeoluwa Matthews, Aninda Manocha, Davide Giri, Marcelo Orenes-Vera, Esin Tureci, Tyler Sorensen, **Tae Jun Ham**, Juan Luis Aragon, Luca P. Carloni, Margaret Martonosi
IEEE International Symposium on Performance Analysis of Systems and Software ***(ISPASS)***
Acceptance Rate : 25/73 ≈ 34% (• **Nominated for the Best Paper Award**)

[ASPLOS '20] IIU: Specialized Architecture for Inverted Index Search
Jun Heo, Jaeyeon Won, Yejin Lee, Shivam Bharuka, Jaeyoung Jang, **Tae Jun Ham**, Jae W. Lee
The 25th ACM International Conference on Architectural Support for Programming Languages and Operating Systems ***(ASPLOS)***
Acceptance Rate : 86/476 ≈ 18%

[HPCA '20] A^3: Accelerating Attention Mechanisms in Neural Networks with Approximation
Tae Jun Ham, Sung Jun Jung, Seonghak Kim, Young H. Oh, Yeonhong Park, Yoonho Song, Jung-Hun Park, Sanghee Lee, Kyoung Park, Jae W. Lee, Deog-Kyoon Jeong
The 26th IEEE International Symposium on High Performance Computer Architecture ***(HPCA)***
Acceptance Rate : 48/235 ≈ 20%

[MICRO '19] Charon: Specialized Near-Memory Processing Architecture for Clearing Dead Objects in Memory
Jaeyoung Jang, Jun Heo, Yejin Lee, Jaeyeon Won, Seonghak Kim, Sung Jun Jung, Hakbeom Jang, **Tae Jun Ham**, Jae W. Lee
*The 52nd IEEE/ACM International Symposium on Microarchitecture (**MICRO**)*
Acceptance Rate : 79/343 ≈ 23%

[IEEE Micro '19] SSDStreamer: Specializing I/O Stack for Large-Scale Machine Learning
Jonghyun Bae, Hakbeom Jang, Jeonghun Gong, Wenjing Jin, Shine Kim, Jaeyoung Jang, **Tae Jun Ham**, Jinkyu Jeong, Jae W. Lee
IEEE Micro, September 2019

[ATC '19] Asynchronous I/O Stack: A Low-latency Kernel I/O Stack for Ultra-Low Latency SSDs
Gyusun Lee, Seokha Shin, Wonsuk Song, **Tae Jun Ham**, Jae W. Lee, Jinkyu Jeong
*USENIX Annual Technical Conference (**ATC**)*
Acceptance Rate : 71/356 ≈ 20%

[ATC '19] Practical Erase Suspension for Modern Low-latency SSDs
Shine Kim, Jonghyun Bae, Hakbeom Jang, Wenjing Jin, Jeonghun Gong, Seungyeon Lee, **Tae Jun Ham**, Jae W. Lee
*USENIX Annual Technical Conference (**ATC**)*
Acceptance Rate : 71/356 ≈ 20%

[ACM TACO '19] Efficient Data Supply for Parallel Heterogeneous Architectures
Tae Jun Ham, Juan L Aragon, Margaret Martonosi
*ACM Transactions on Architecture and Code Optimization (**TACO**)*
Presented on **HiPEAC** 2020 Conference

[ACM TACO '17] Decoupling Data Supply from Computation for Latency-Tolerant Communication in Heterogeneous Architectures
Tae Jun Ham, Juan L Aragon, Margaret Martonosi
*ACM Transactions on Architecture and Code Optimization (**TACO**)*

[MICRO '16] Graphicionado: A High-Performance and Energy-Efficient Accelerator for Graph Analytics
Tae Jun Ham, Lisa Wu, Narayanan Sundaram, Nadathur Satish, Margaret Martonosi
*The 49th IEEE/ACM International Symposium on Microarchitecture (**MICRO**)*
Acceptance Rate : 61/283 ≈ 22% (• MICRO-49 Best Paper Award)

[MICRO '15] DeSC: Decoupled Supply-Compute Communication Management for Heterogeneous Architectures
Tae Jun Ham, Juan L Aragon, Margaret Martonosi
*The 48th IEEE/ACM International Symposium on Microarchitecture (**MICRO**)*
Acceptance Rate : 61/283 ≈ 22% (• IEEE Micro's Top Picks Honorable Mention)

[HPCA '13] Disintegrated Control for Energy-Efficient and Heterogeneous Memory Systems
Tae Jun Ham, Bharath K. Chelepalli, Neng Xue, Benjamin C. Lee
*The 19th IEEE International Symposium on High Performance Computer Architecture (**HPCA**)*
Acceptance Rate : 51/249 ≈ 20%

Patents

- Apparatus and method with scheduling (US Patent 12,524,267)
  with Jae Wook Lee, Younghwan Oh, and Yunho Jin
- Scheduler, method of operating the same, and accelerator apparatus including the same (US Patent 12,277,440)
  with Seung Wook Lee, Jae Wook Lee, Young Hwan Oh, and Seng Hak Kim
- Processor, method of operating the processor, and electronic device including the same (US Patent 12,327,179)
  with Seung Wook Lee, Jaeyeon Won, and Jae Wook Lee
- Layer-wise scheduling on models based on idle times (US Patent 12,099,869)
  with Seung Wook Lee, Younghwan Oh, Jaewook Lee, Sam Son, and Yunho Jin
- Hardware accelerator performing search using inverted index structure and search system including the hardware accelerator (US Patent 11,544,270)

with Jun Heo, Jaeyeon Won, Yejin Lee, Jaeyoung Jang, and Jae Wook Lee

- **Hammer refresh row address detector, and semiconductor memory device and memory module including the same** (US Patent 11,568,917)
  with Hoon Shin, Yeonhong Park, Jaewook Lee, Eojin Lee, Woosuk Kwon, and Jungho Ahn
- **Method for candidate selection and accelerator for performing candidate selection** (US Patent 11,636,173)
  with Seonghak Kim, Sungjun Jung, Younghwan Oh, Jaewook Lee, Deog-Kyoon Jeong, and Minsoo Lim
- **Device for accelerating self-attention operation in neural networks** (US Patent App. 17/864,235)
  with Ye Jin Lee, Seong Hoon Seo, Soo Sung Kim, Hyun Ji Choi, Jae W. Lee, and Sung Jun Jung
- **Accelerator system for training deep neural network model using nand flash memory and operating method thereof** (US Patent App. 18/089,141)
  with Jae W. Lee, Yunho Jin, Jong Hyun Bae, and Gin A Sohn
- **Method for processing page fault by processor** (US Patent 11,436,150)
  with Jinkyu Jeong, Jae Wook Lee, and Wenjing Jin
- **Electronic device and method with scheduling** (US Patent App. 17/195,748)
  with Seung Wook Lee, Younghwan Oh, Jaewook Lee, Sam Son, and Yunho Jin
- **Instruction, Circuits, and Logic for Graph Analytics Acceleration** (US Patent App. 15/089,232)
  with Lisa K. Wu, Narayanan Rajagopalan Satish, and Narayanan Sundaram

Professional Service

- **MICRO**: Program Committee (2026), External Review Committee (2025, 2024), Student Research Competition Selection Committee (2023)
- **HPCA**: Program Committee (2025, 2026), Industry Track Program Committee (2026)
- **ISCA**: Program Committee (2025), External Review Committee (2024)
- **ASPLOS**: Program Committee (2023, 2024), External Review Committee (2019)
- **IISWC**: Program Committee (2024)
- **YArch**: Program Committee (2024)
- **IEEE Micro Top Picks**: Program Committee (2023)
- **CGO**: Web Chair (2021, 2022)

Paper Reviews

- IEEE Transactions on Computers (TC), IEEE TVLSI, IEEE TMC, IEEE TCAD
- IEEE Computer Architecture Letters (CAL), IEEE Micro
- ACM Transactions on Architecture and Code Optimization (TACO)
- ACM Transactions on Parallel Computing (TOPC)
- Elsevier Future Generation Computer Systems (FGCS)

Invited Talks

- KAIST, POSTECH (Sep 2017)
  DeSC: Decoupling Data Supply from Computation for Latency-Tolerant Communication
- DARPA HIVE PI Meeting (Oct 2017)
  Graphicionado: A High-Performance and Energy-Efficient Accelerator for Graph Analytics
- HiPEAC 2020 (Jan 2020)
  Efficient Data Supply for Parallel Heterogeneous Architectures
- Seoul National University AI Summer School (Aug 2020)
  Accelerating Neural Network Attention Mechanism with HW/SW Codesign
- POSTECH Summer AI Seminar (Aug 2020)
  Hardware/Software Co-design for Modern AI and Data Analytics Applications

Research Mentoring

During my tenure as a postdoctoral researcher at Seoul National University, I closely mentored and collaborated with the following students (in partnership with Prof. Jae W. Lee), overseeing research projects from initial concept through system implementation to high-impact publication.

Graduate Students

- **Jaeyoung Jang** (Ph.D., SKKU → Samsung Electronics)
- **Young H. Oh** (Ph.D., SKKU → Samsung Electronics → Ajou University)
- **Jun Heo** (Ph.D., SNU → Samsung Electronics → MangoBoost)
- **Jonghyun Bae** (Ph.D., SNU → LBNL → Google Visiting Researcher → HyperAccel)
- **Shine Kim** (Ph.D., SNU → Samsung Electronics)
- **Seung Yul Lee** (Ph.D., SNU → SNU Postdoc)
- **Yeonhong Park** (Ph.D., SNU → Meta)
- **Wenjing Jin** (Ph.D. student, SNU → Samsung Electronics)
- **Sung Jun Jung** (Ph.D., SNU → SK Hynix)
- **Yejin Lee** (Ph.D., SNU → Meta)
- **Seong Hoon Seo** (Ph.D., SNU)
- **Soosung Kim** (Ph.D., SNU)
- **Jeonghun Gong** (M.S., SNU → Samsung Electronics)
- **Yunho Jin** (M.S., SNU → Harvard Ph.D. → Google)
- **Sam Son** (M.S., SNU → UC Berkeley Ph.D. → Meta)
- **Hyunji Choi** (M.S., SNU → Meta)

Undergraduate Students

- **Jaeyeon Won** (B.S., SNU → MIT Ph.D.)
- **Stuart Sul** (B.S., SNU → Blux → Stanford M.S. → Cursor)
- **U Gyeong Song** (B.S., SNU)

Technical Skills

- **Datacenter Performance & Efficiency**: Fleet-wide resource optimization, performance engineering for distributed systems, capacity efficiency, and infrastructure-scale bottleneck analysis.
- **Hardware/Software Co-Design**: Cross-stack optimization bridging distributed systems and silicon microarchitecture; evaluation, optimization, and architectural guidance for future SoC and server platforms.
- **Distributed Database Systems**: Architecture and optimization of large-scale distributed databases and storage systems; expertise in foundational structures including LSM-trees and Key-Value stores.
- **ML for Systems & Systems for ML**: Leveraging machine learning for system performance, efficiency, and co-design, alongside infrastructure-scale performance optimization for machine learning workloads.
- **Modeling, Simulation & Profiling**: Performance modeling, full-stack system simulation, and architectural evaluation frameworks; **Google-scale production profiling infrastructure** and custom system-level analysis tools.
- **Technical Leadership & Management**: Former Tech Lead Manager (TLM) for a 20-person team at Google; expertise in technical strategy, organizational roadmapping, and research mentorship (Ph.D. level).
- **Implementation**: High-performance C/C++, Python, LLVM, and large-scale infrastructure codebase management.